

Lab 10

Noise

PART 1: LPF AC Noise Analysis

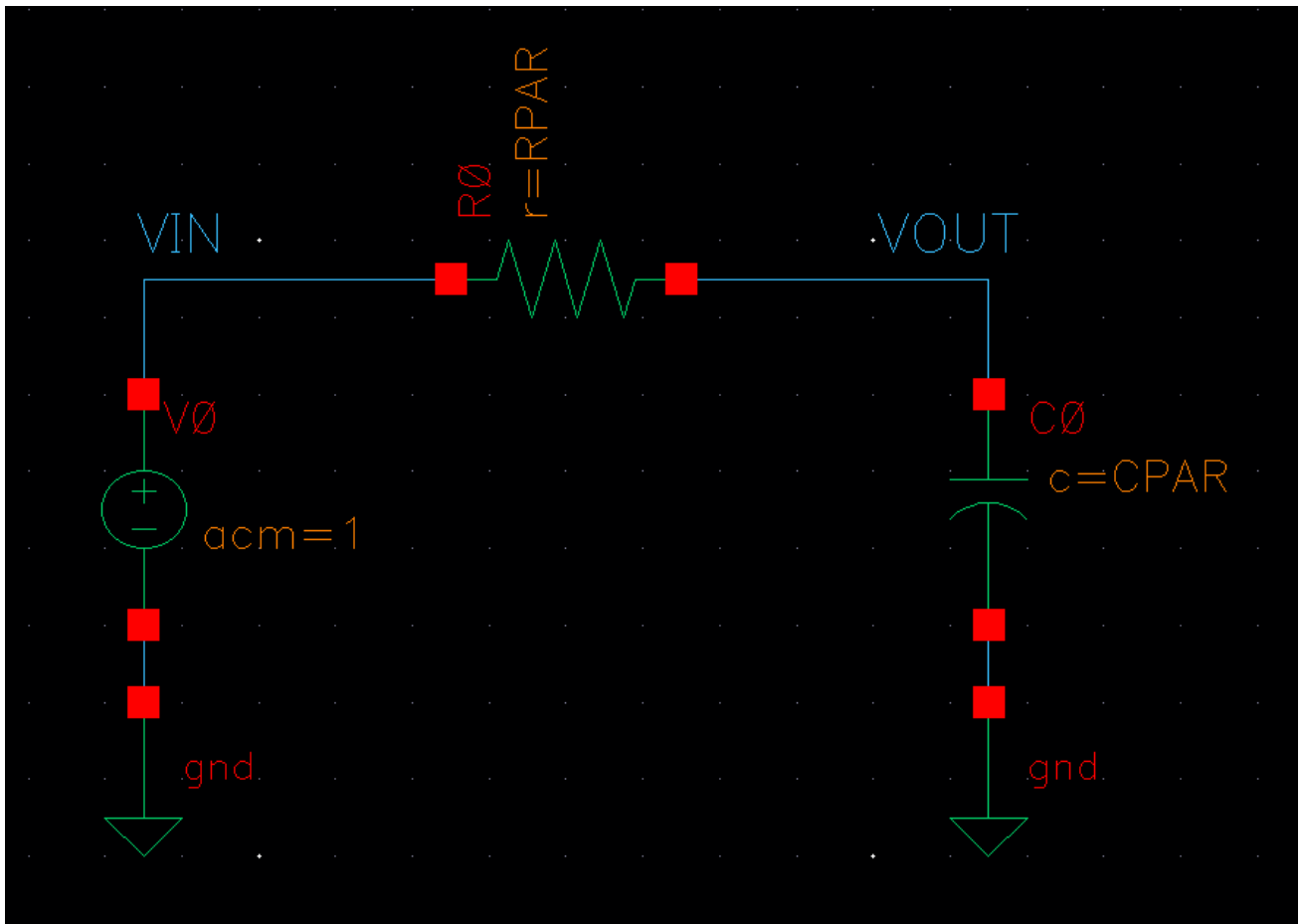


Figure 1 RC LPF Filter

```

1 .control
2     save all
3     shell rm lab10.raw
4     noise V(VOUT) V1 DEC 10 1 10g
5     save onoise_total
6     print onoise_total
7     setplot noisel
8     plot onoise_spectrum
9     plot inoise_spectrum
10
11     save all
12     write lab10.raw
13 .endc

```

Figure 2 Noise Analysis Code

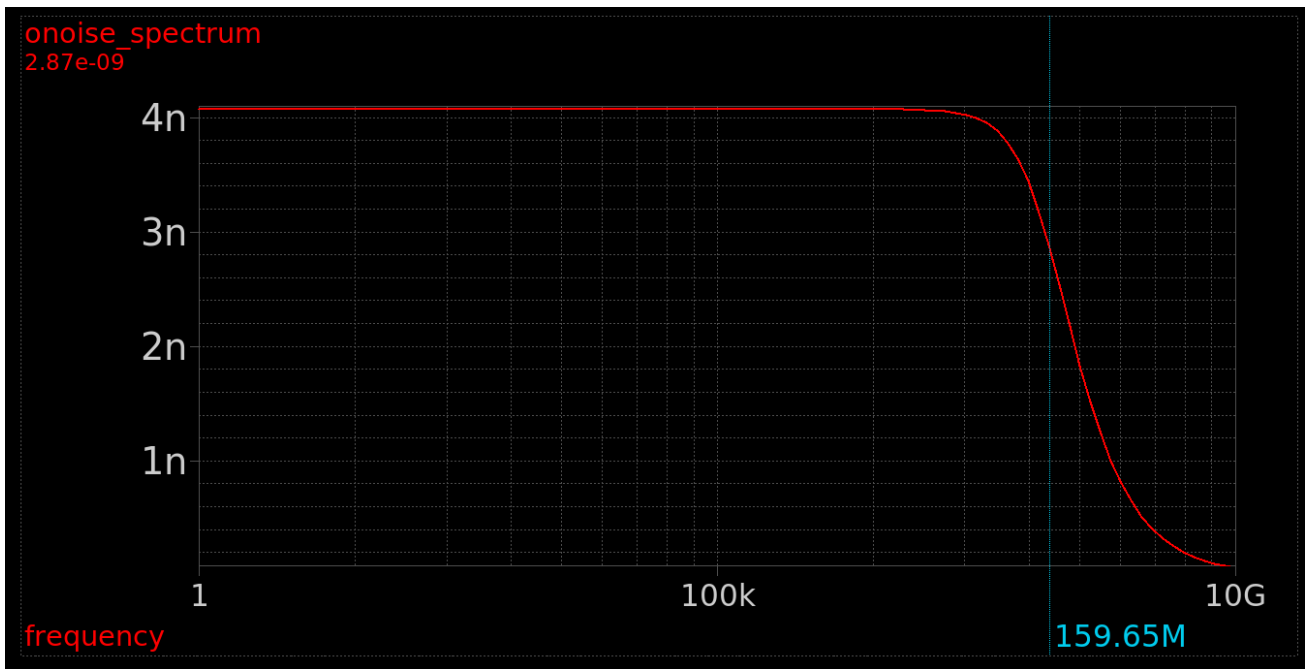


Figure 3 Noise Spectrum BW Annotated

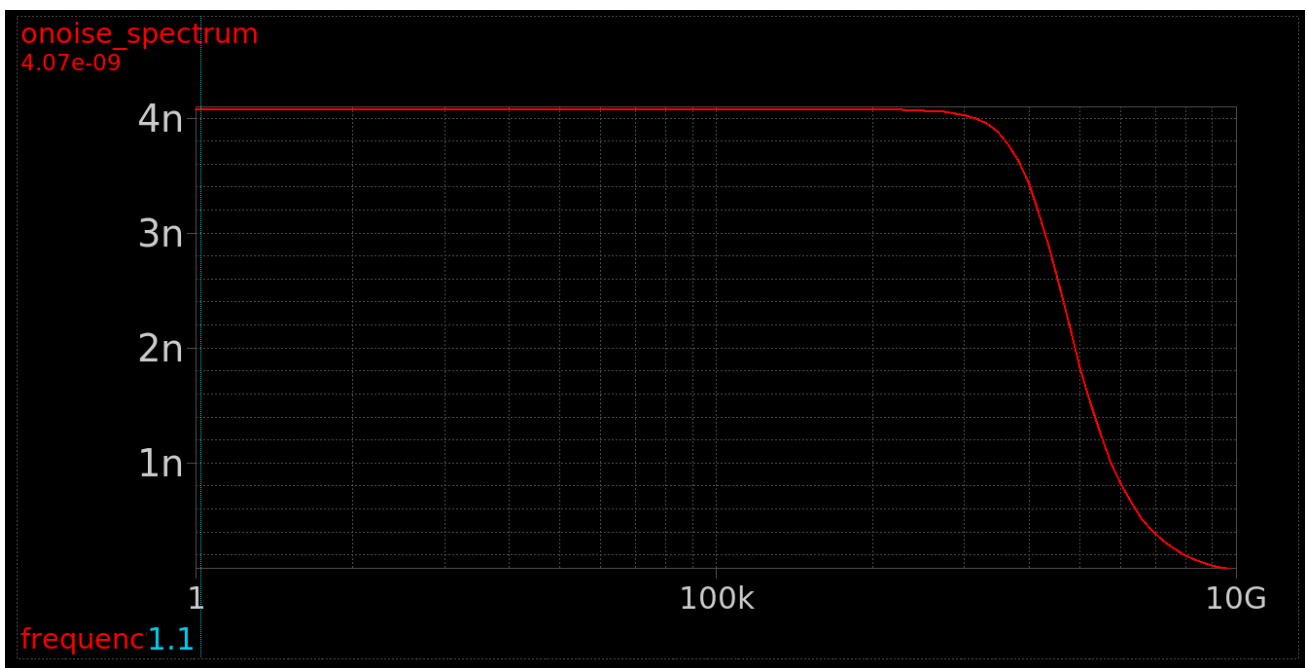


Figure 4 Output Noise Annotated

$$\text{onoise_total} = 6.397581\text{e-}05$$

Figure 5 Total Output Noise RMS Value from xschem

Hand Analysis:

$$V_n = \sqrt{4kTR} = \frac{4.07nV}{\sqrt{Hz}}$$

$$BW = \frac{1}{2\pi RC} = 159.155MHz$$

$$V_{nourms} \approx \sqrt{\frac{1p}{C}} * 64\mu = 64\mu$$

	Simulation	Hand Analysis
Vn	4.071n	4.07n
BW	159.65MHz	159.155MHz
Vnourms	63.97u	64u

Simulation and Hand Analysis Results agree with each other

- Run parametric sweep for RPAR = 1k, 10k, 100k, 1000k

getData("/out" ?result "noise")

Sun Aug 24 07:31:52 2025 1

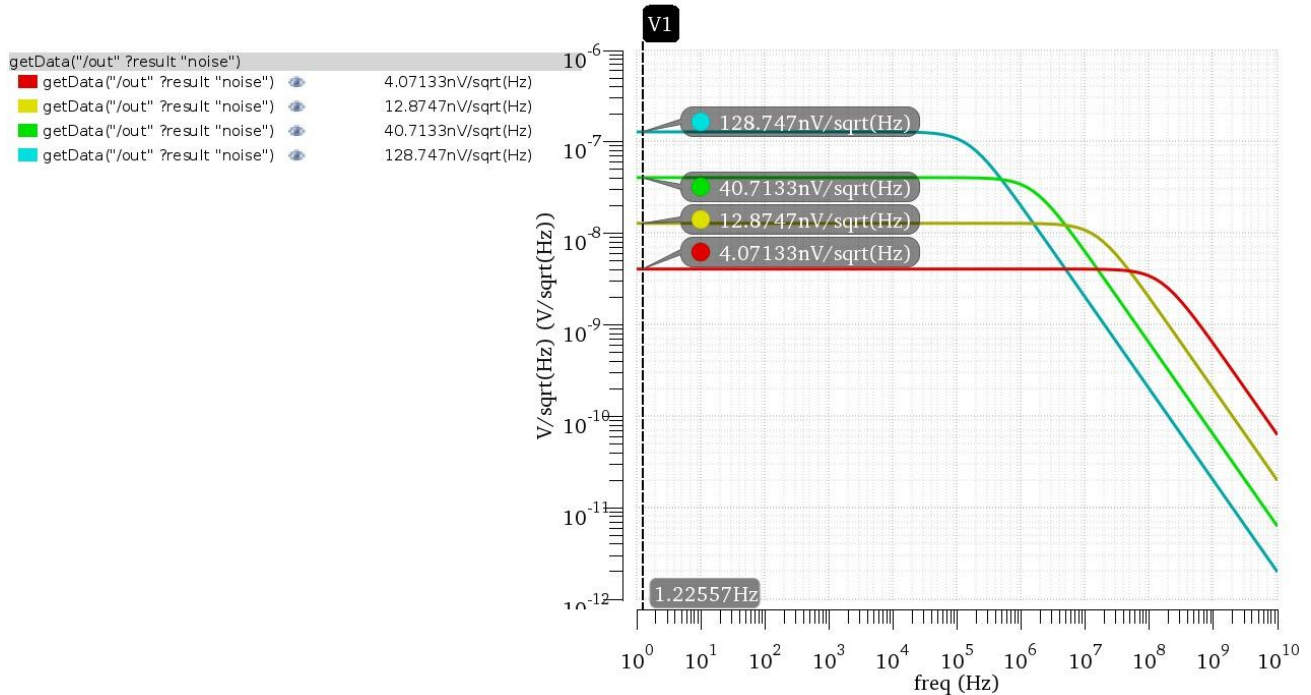


Figure 6 Output Noise results Overlaid

As we increase the Value of RPAR the bandwidth decreases by a factor of 10 and noise increases by a factor of root10

RPAR=1k		
Lab10_ITI:noise_rc:1	rmsNoise(1 1e+10)	64.32u
Lab10_ITI:noise_rc:1	bandwidth(getData("/out" ?re...	158.8M
Lab10_ITI:noise_rc:1	getData("/out" ?result "noise")	
RPAR=10k		
Lab10_ITI:noise_rc:1	rmsNoise(1 1e+10)	64.62u
Lab10_ITI:noise_rc:1	bandwidth(getData("/out" ?re...	15.88M
Lab10_ITI:noise_rc:1	getData("/out" ?result "noise")	
RPAR=100k		
Lab10_ITI:noise_rc:1	rmsNoise(1 1e+10)	64.65u
Lab10_ITI:noise_rc:1	bandwidth(getData("/out" ?re...	1.588M
Lab10_ITI:noise_rc:1	getData("/out" ?result "noise")	
RPAR=1M		
Lab10_ITI:noise_rc:1	rmsNoise(1 1e+10)	64.66u
Lab10_ITI:noise_rc:1	bandwidth(getData("/out" ?re...	158.8k
Lab10_ITI:noise_rc:1	getData("/out" ?result "noise")	

Figure 7 RMS and Bandwidth values for each run

The RMS value of Noise is constant as it doesn't depend on the value of R

PART 2: LPF Transient Noise Analysis

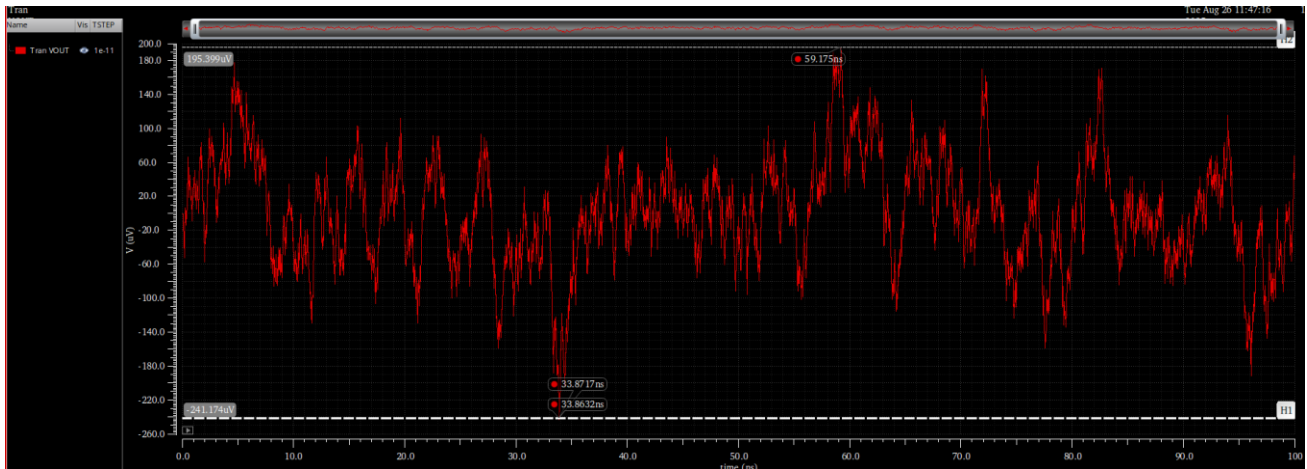
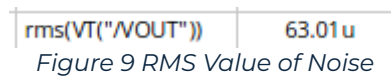


Figure 8 Noise output with min and max annotated



The RMS Value of noise is close the one calculated in the previous part.

Repeating simulation with Tstep = TAU/10

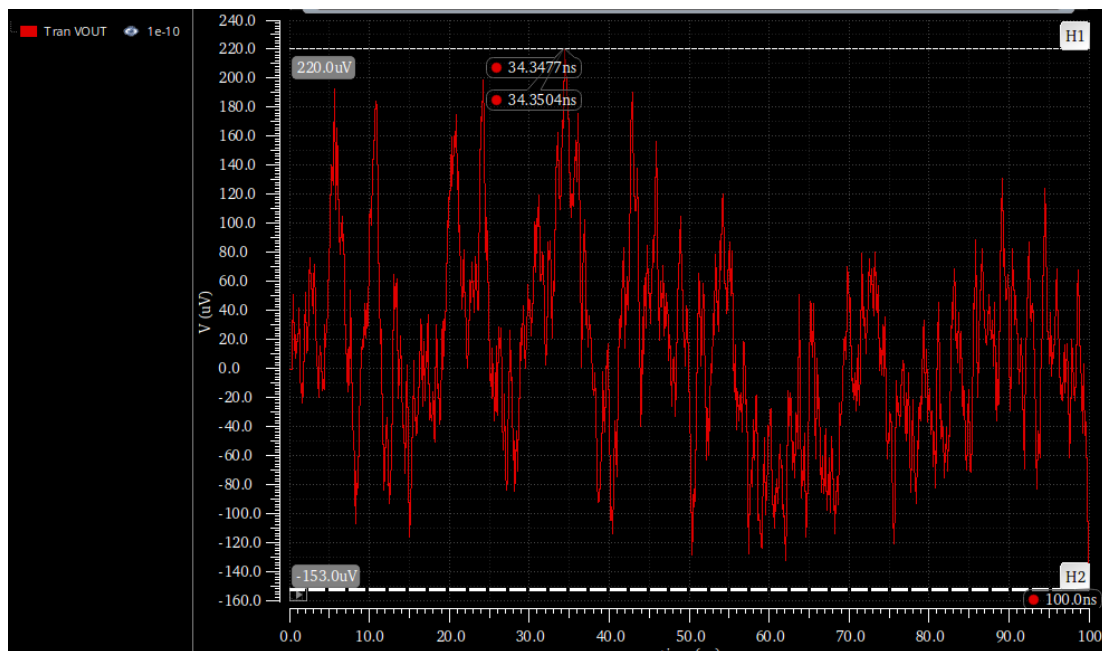
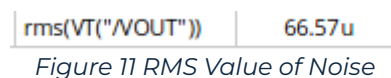


Figure 10 Output Noise annotated



Output Noise RMS Increased Slightly as Fmax increased so the area integrated is bigger.

Running 20 Runs:

rms(VT("/VOUT"))

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Name ...
rms(VT("/VOUT"))

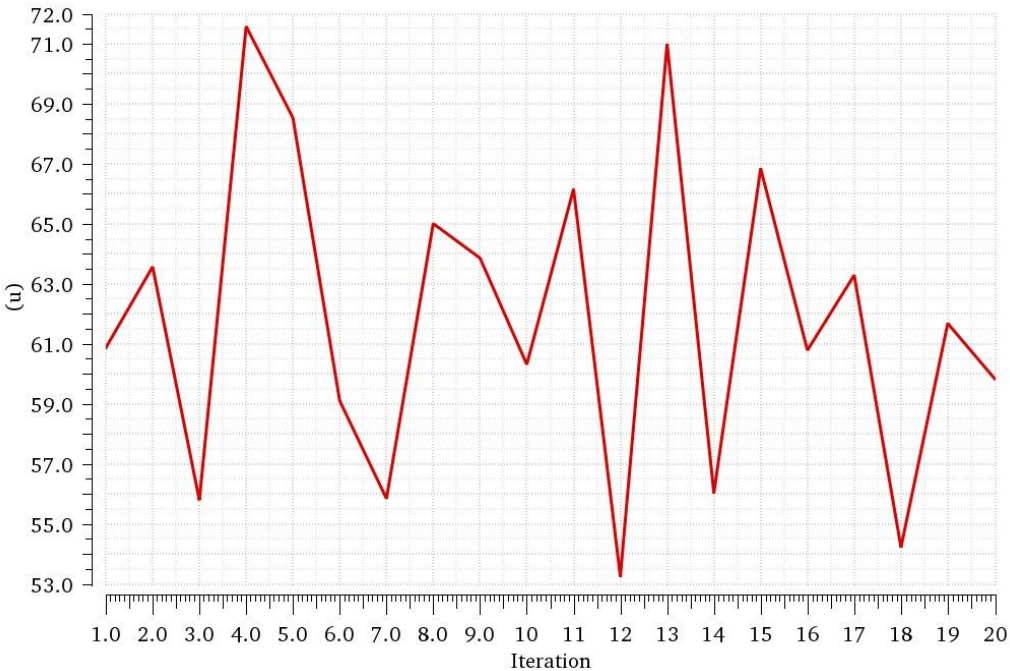


Figure 12 RMS Noise vs Iterations

average(rms(VT("/VOUT")))	61.97u
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Figure 13 Average Noise RMS Value

The value is slightly smaller due to the different extremes from each iteration but still comparable to what was calculated in parts 1 and 2.

PART 3: 5T OTA AC Noise Analysis

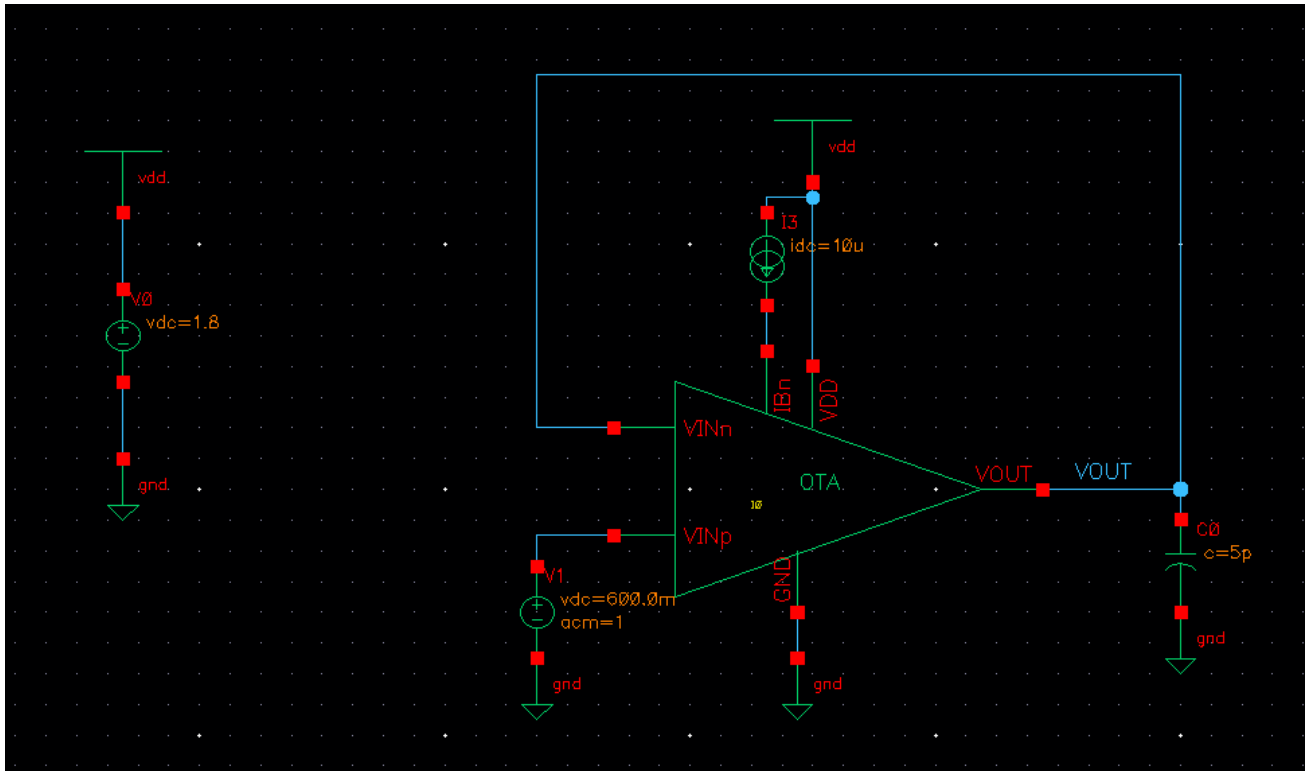


Figure 14 OTA Unity Buffer Schematic

BSIM4v5: Berkeley Short Channel IGFET Model-4			
device	m.x1.xm6.m0	m.x1.xm5.m0	m.x1.xm3.m0
model	nmos_3p3.10	nmos_3p3.14	pmos_3p3.8
id	1e-05	3.91602e-05	1.94154e-05
gm	9.78173e-05	0.000349102	0.000116439
gmbs	3.84057e-05	0.000137761	4.41489e-05
gds	1.92449e-07	6.73433e-05	2.28334e-06
vgs	0.822205	0.822205	1.04218
vth	0.667182	0.668168	0.756427
vds	0.822204	0.144526	0.892294
vdsat	0.161448	0.161665	0.286699
BSIM4v5: Berkeley Short Channel IGFET Model-4			
device	m.x1.xm4.m0	m.x1.xm2.m0	m.x1.xm1.m0
model	pmos_3p3.8	nmos_3p3.12	nmos_3p3.12
id	1.97448e-05	1.94154e-05	1.97448e-05
gm	0.000118099	0.00032091	0.00032495
gmbs	4.47105e-05	9.98597e-05	0.000101132
gds	2.12122e-06	2.67513e-06	2.89281e-06
vgs	1.04218	0.763162	0.76547
vth	0.755461	0.752227	0.752587
vds	1.04217	0.763161	0.613285
vdsat	0.28748	0.0935059	0.0944818

Figure 15 OTA OP Point in buffer config for Hand Analysis

Thermal Noise Calculations:

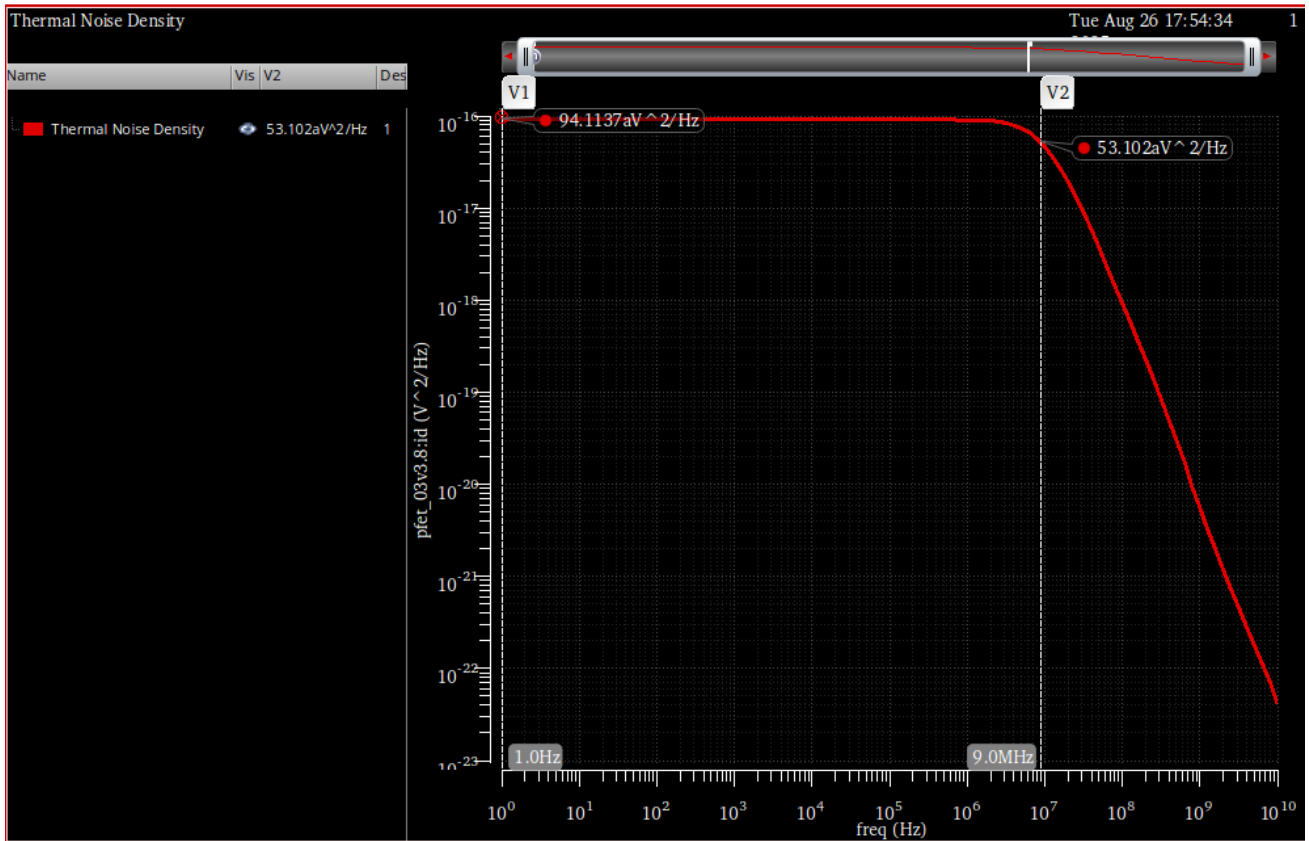


Figure 16 Thermal Noise Output Density

$$V_{nout}^2 = \frac{94.1 \text{ aV}^2}{\text{Hz}}$$

Hand Analysis:

$$V_n^2 = \frac{8kT\gamma}{gm_{1,2}} \left(1 + \frac{gm_{3,4}}{gm_{1,2}} \right) \approx \frac{90.825 \text{ aV}^2}{\text{Hz}}$$

RMS Value of Thermal Noise:

/I0/M0	id	5.35563e-10
/I0/M1	id	5.33758e-10
/I0/M4	id	2.29182e-10
/I0/M5	id	2.01998e-10
/I0/M2	id	3.43088e-12
/I0/M3	id	1.83001e-12

Figure 17 Thermal Noise of Each Transistor from Noise Summary

$$V_{n \text{ thermal rms}}^2 \approx \frac{1.4993nV^2}{Hz}$$

Noise Bandwidth:

Thermal Noise BW	10.19M
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Figure 18 Amplifier Noise BW

Hand Analysis:

$$BW_{nois} = BW_{amp} = 10MHz$$

$$V_{n \text{ thermal rms}}^2 = V_n^2 * B_n = 90.825a * \frac{\pi}{2} 10M = \frac{1.4266nV^2}{Hz}$$

	Simulation	Hand Analysis
V_n Thermal	94.1a	90.825a
V_n Thermal RMS	1.499n	1.4266n
BW	10.19MHz	10MHz

Hand Analysis and simulation results agree with each other.

Total Output Noise and Noise Corner:

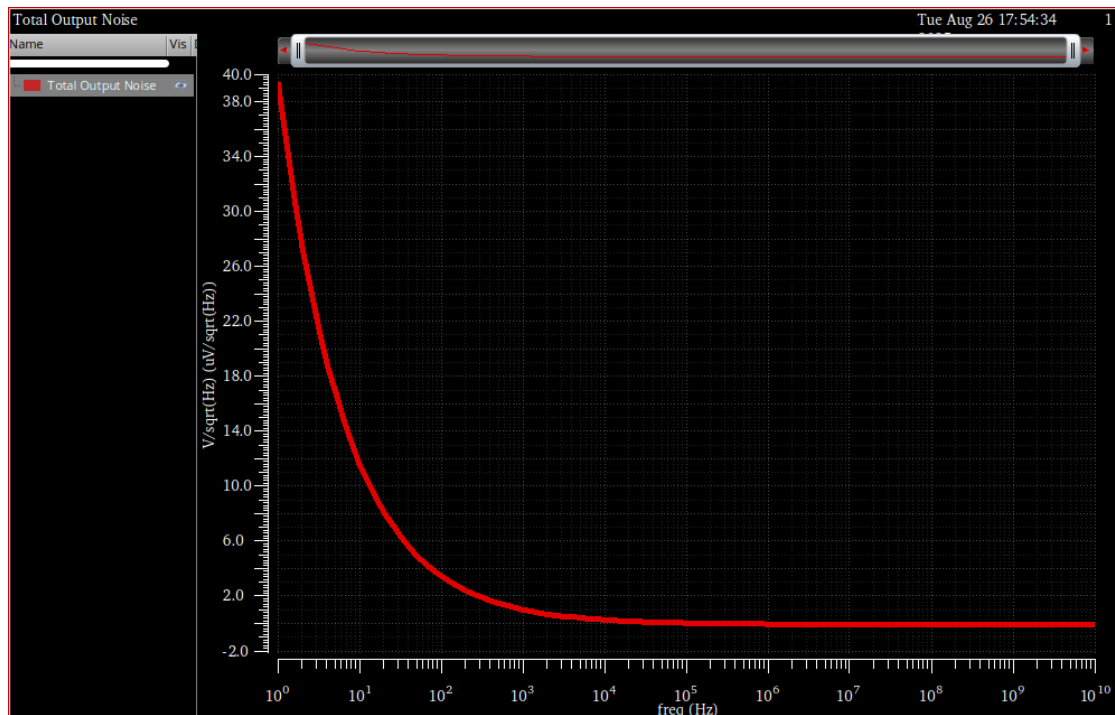


Figure 19 Total Output Noise Density

Estimating Noise Corner:

To estimate the noise corner I plotted both the total Thermal noise and Flicker noise spectrums and estimated where they both cross paths. I changed the yscale to dB to better visualize it

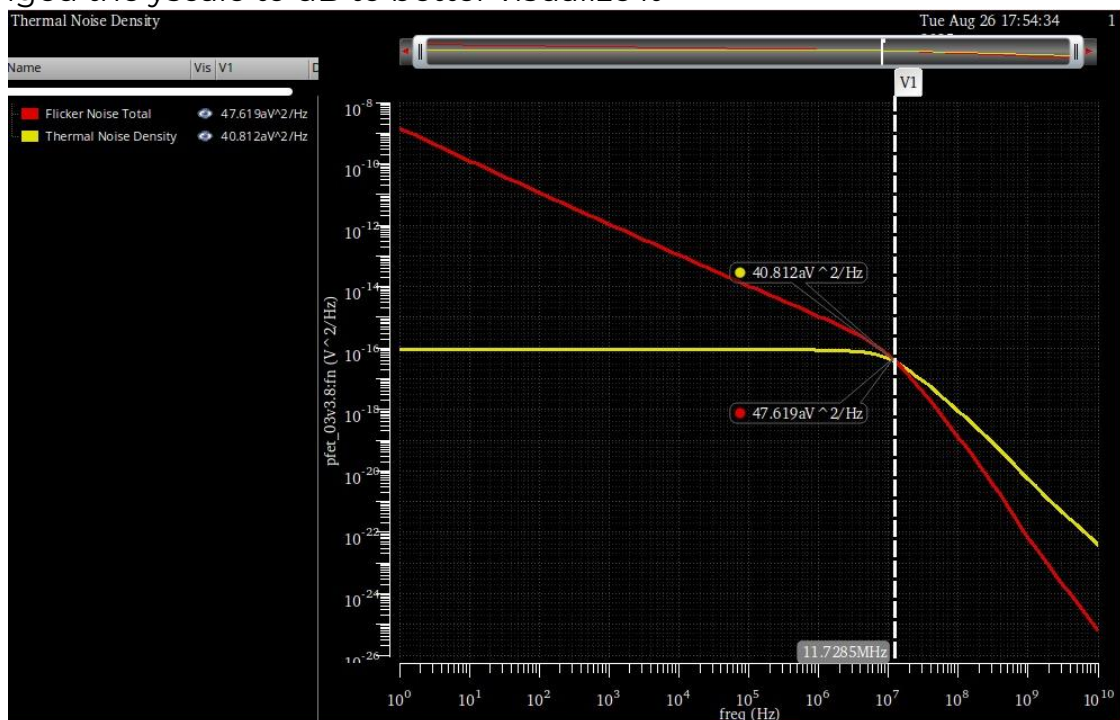


Figure 20 Estimate of Noise Corner

Noise Corner $\approx 11.728\text{MHz}$

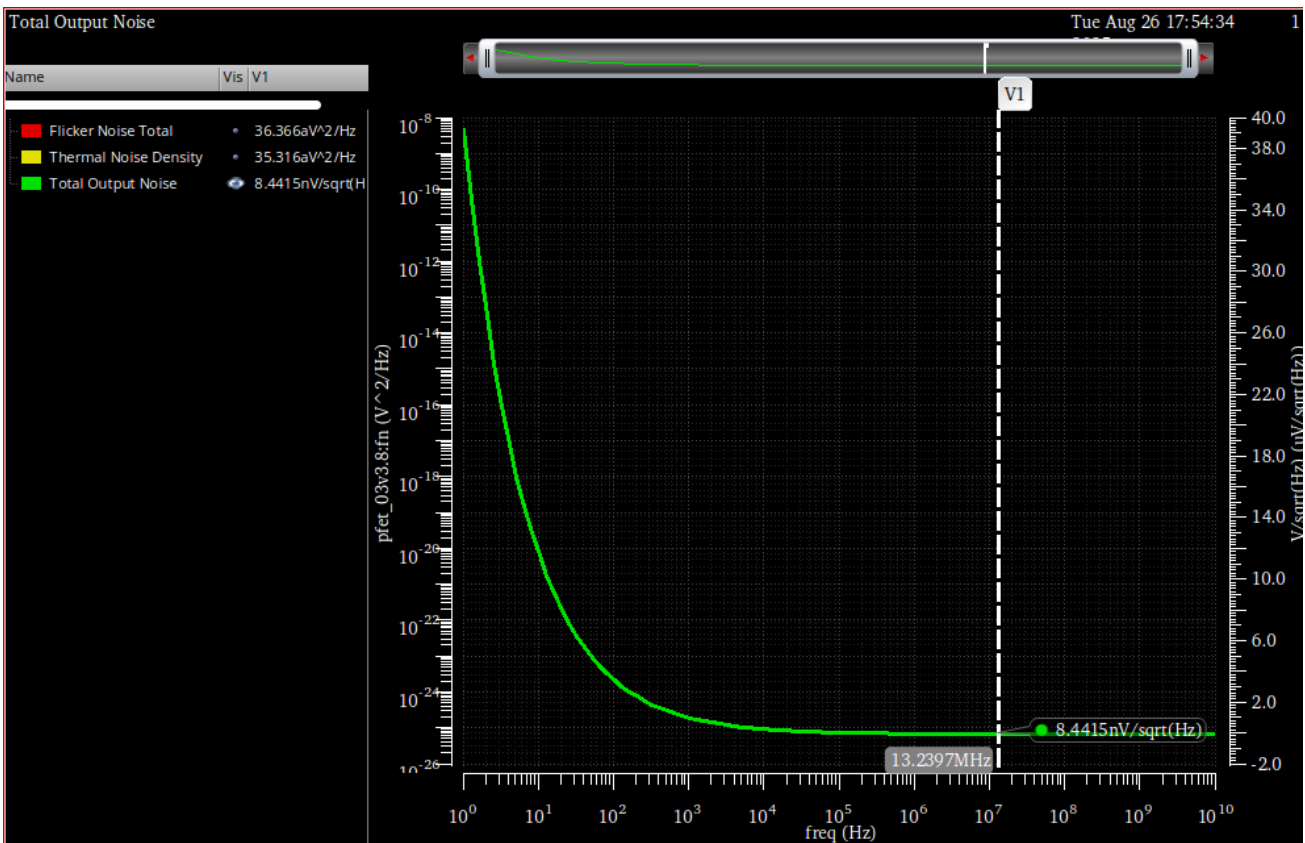


Figure 21 Noise Corner annotated for in total Noise spectrum

PART 4: 5T OTA Transient Noise Analysis

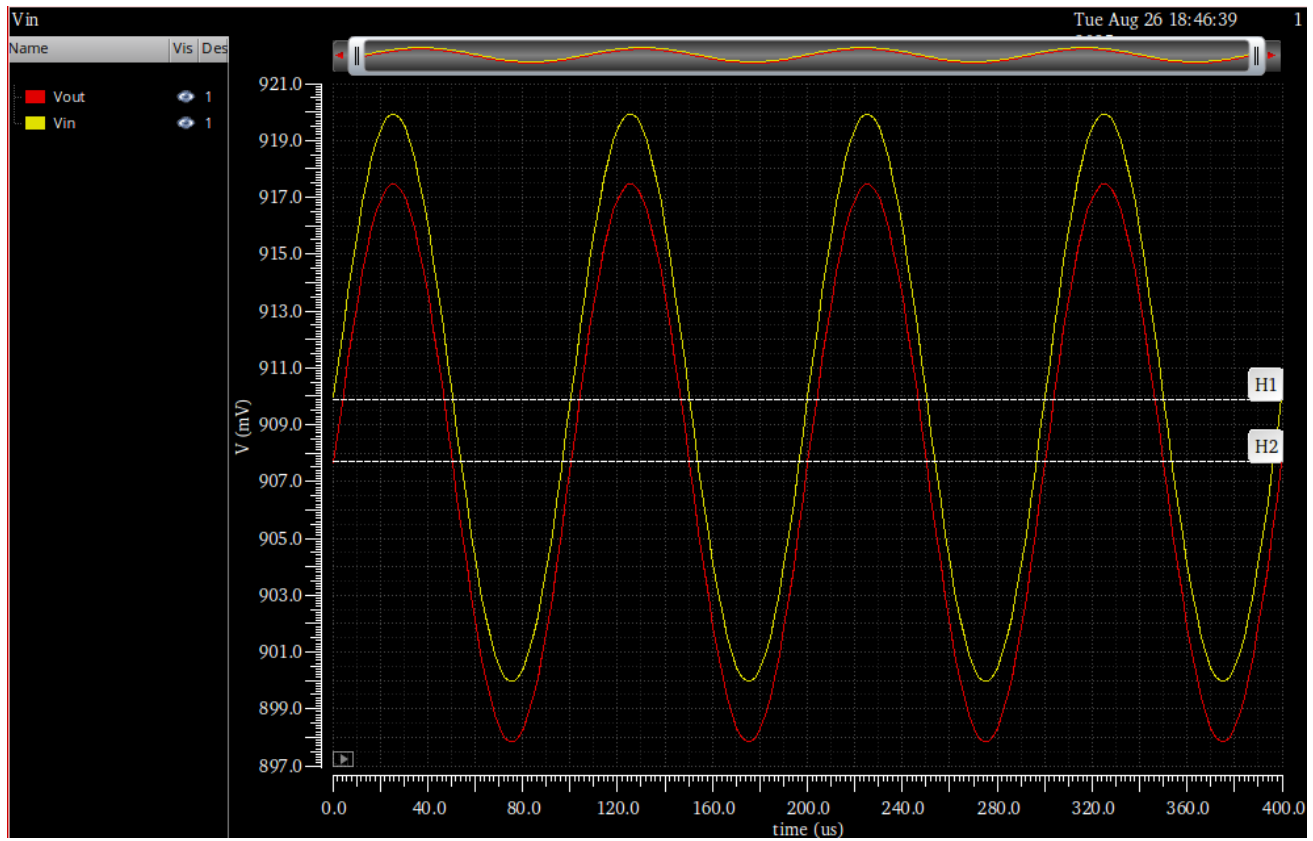


Figure 22 V_{in} and V_{out} Overlaid

Both V_{in} and V_{out} match pretty well with just a slight difference in the DC Level due to the lower gain of the amplifier resulting in higher v_{err} thus a bigger difference between the output and input voltages.

Noisy Transient Simulation:

☒ Transient Noise

Noise Fmax

Noise Fmin

Noise Seed

Noise Scale

Figure 23 Upper Noise frequency at 10 times the amp GBW

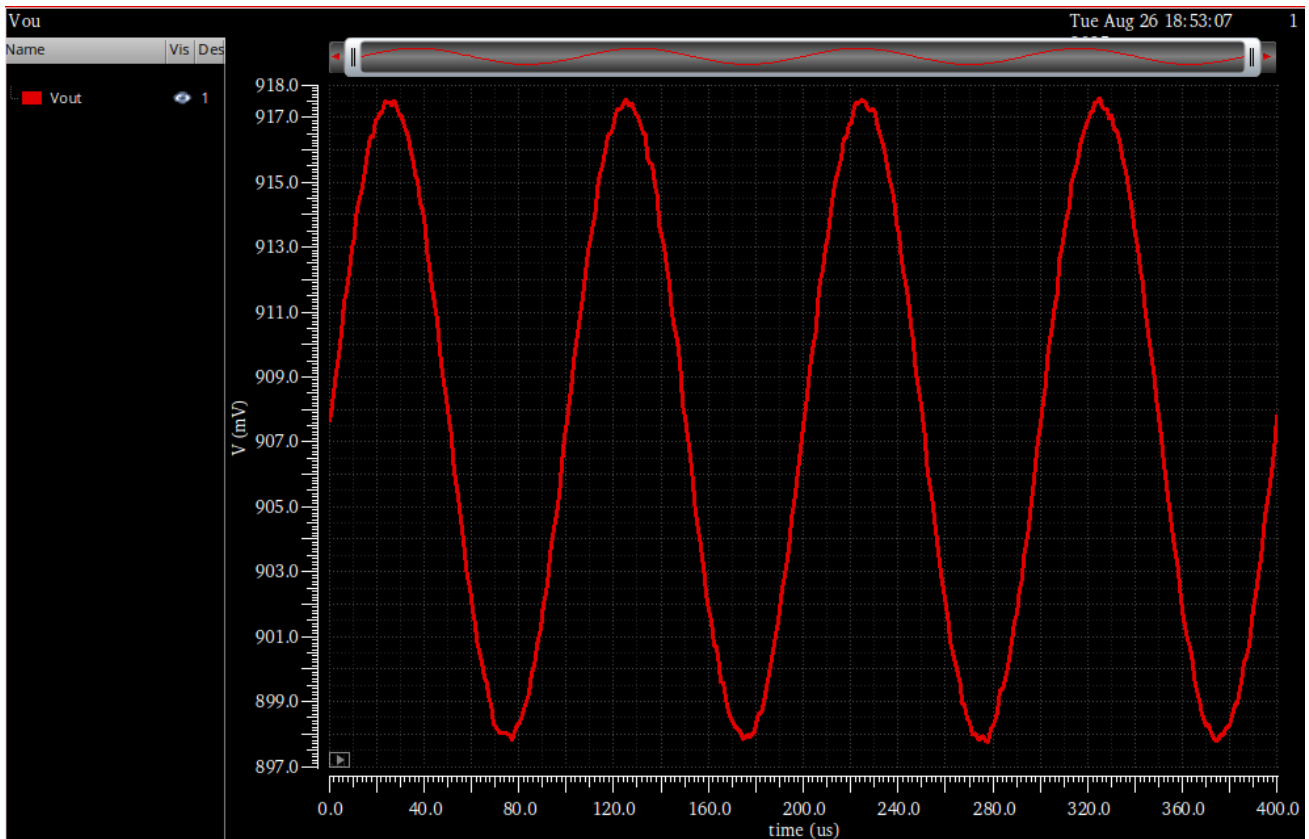


Figure 24 Noisy Output

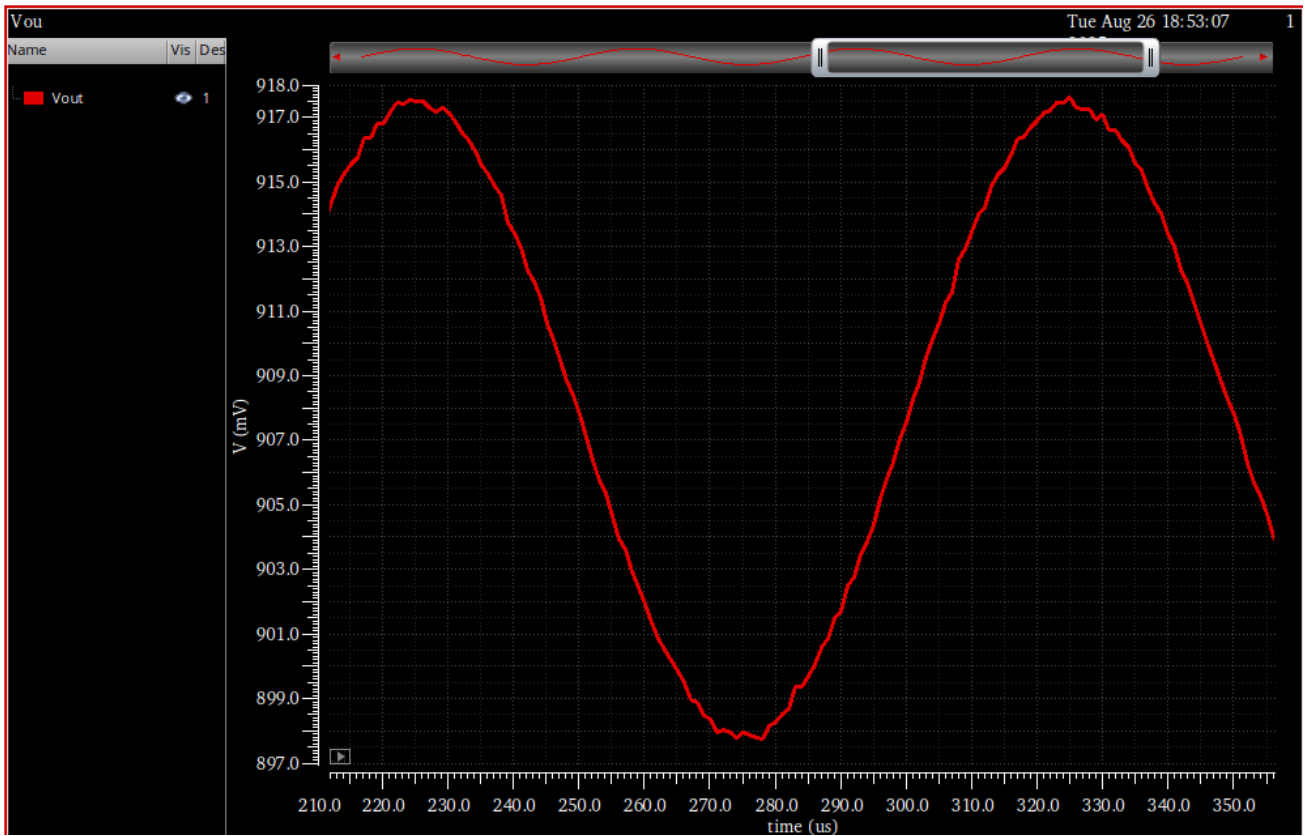


Figure 25 Noise Up close

20 Noise Iterations:

To Isolate the noise from the output I used this expression to use the clean run and subtract the signal from the noise:

```
Output Noise RMS    expr    rms((v("/VOUT" ?result "tran") - value(v("/VOUT" ?result "tran" ?resultsDir "/home/centos/simulation/Lab10_ITI/noise_mos/adexl/results/data/Clean_Tran/psf/Lab10_ITI:noise_mos:1/psf") "Design_Point" 1)))
```

Figure 26 Output Setup for output noise RMS

```
rms((v("/VOUT" ?result "tran") - value(v("/VOUT" ?result "tran" ?resultsDir
"/home/centos/simulation/Lab10_ITI/noise_mos/adexl/results/data/Clean_Tran/psf/Lab10_ITI:noise_mos:1/psf") "Design_Point" 1)))
```

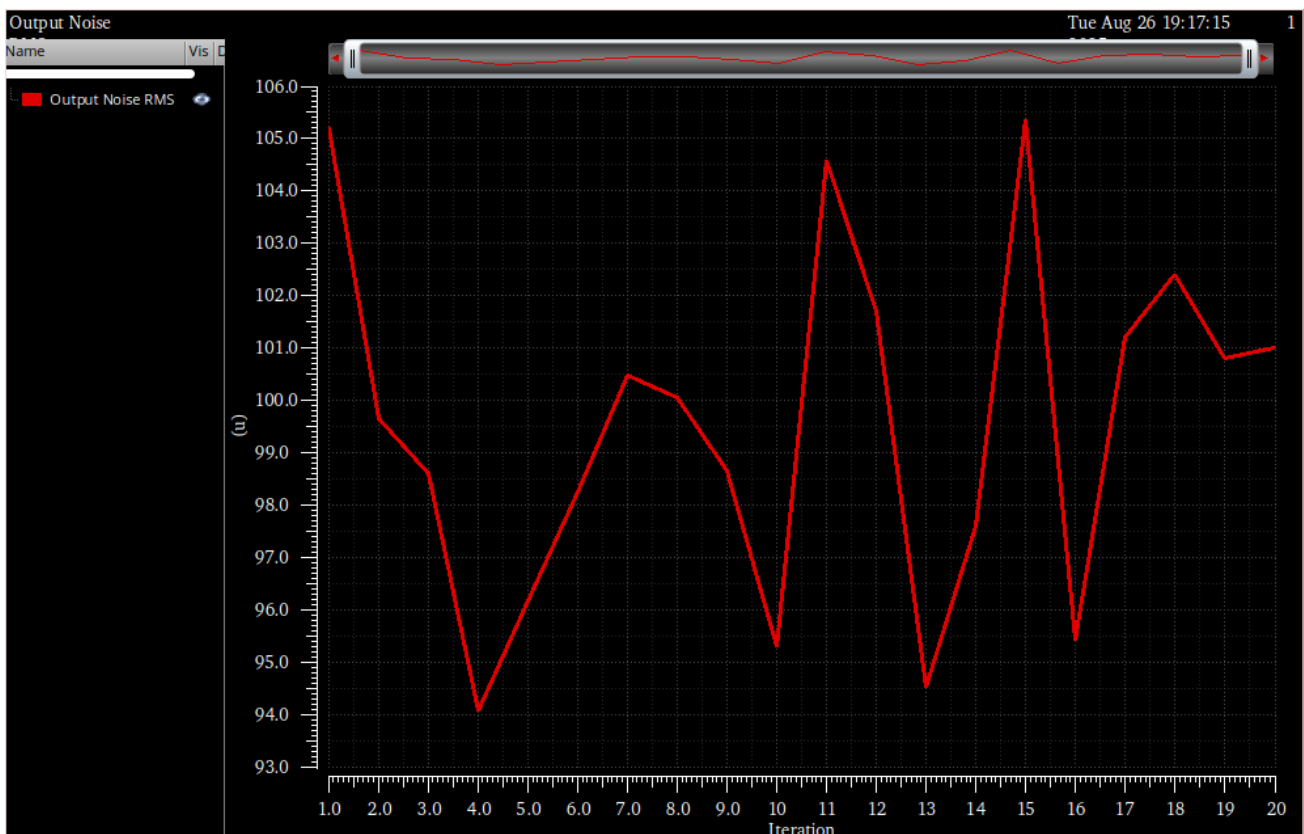


Figure 27 Output Noise vs Iterations

```
average(rms((v("/VOUT" ?result ... 99.39u
```

Figure 28 Average Noise RMS

Compared to Previous part:

The value is different as the value from the previous part calculated the average RMS for the squared voltage, but here it is non squared.

Furthermore, it would be slightly higher than the previous part because this simulation takes into consideration the flicker noise which we ignored in hand analysis for simplification.